

Requirements Table — Digital Campus Library Reservation Gateway

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Problem no: 8

Section: H

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow members to search the ISBN repository and place hold requests on currently loaned books to enter a FIFO waitlist.	High	Pass: User is assigned correct position in queue. Fail: User already holding max allowed books places additional reservation.	Enables self-service reservation of unavailable titles; forms the core lending workflow of the library.
FR-002	Functional	The system shall verify a member's total active loans and holds against the maximum permitted (5 items) before confirming any new hold request.	High	Pass: A hold request is rejected with a clear error message when the member's active loan+hold count is already at or above 5. Fail: A hold request is accepted despite the member being at or above the limit.	Enforces the library's fair-borrowing policy and prevents a single member from monopolizing high-demand titles.
FR-003	Functional	The system shall automatically notify the next member in a title's FIFO waitlist by email/SMS within 5 minutes of that title being checked back in, and hold the copy for that member for 48 hours.	High	Pass: A timestamped notification log shows the correct member notified within 5 minutes of the return event, and the 48-hour hold timer starts at that moment. Fail: No notification is sent, the wrong member is notified, or the hold timer does not start.	Automates the holding-notification workflow named in the problem brief and prevents returned copies from sitting idle.
FR-004	Functional	The system shall calculate an overdue fine at a fixed daily rate for any book returned after its due date and update the member's outstanding balance immediately upon check-in.	Medium	Pass: Outstanding balance = (days overdue x daily rate) and is visible on the member's account within the same session as check-in. Fail: The fine is miscalculated or the balance is not updated at check-in.	Gives the library an accurate, auditable record of overdue fine transactions, as required by the problem brief.
FR-005	Functional	The system shall allow the Head Librarian to add, update, or deactivate ISBN catalog entries (title, author, copy count), with changes reflected in member-facing search immediately.	Medium	Pass: A catalog edit persists and appears in search results on the next query. Fail: An edit is lost or is not searchable after saving.	Keeps the catalog members search against accurate and current.

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NFR-001	Performance & Security	The catalog search query shall return paginated results for 50,000 titles in less than 150 ms.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Keeps search responsive for students during peak enrolment/exam periods when catalog traffic is highest.
NFR-002	Reliability & Availability	The system shall maintain at least 99.9% uptime during official library operating hours (8 AM-10 PM) and automatically fail over to a backup notification queue if the primary notification service is unavailable.	Medium	Pass: Monthly monitoring logs show $\leq 0.1\%$ downtime during operating hours, and a simulated primary-service outage shows notifications resuming via the backup queue within 2 minutes. Fail: Downtime exceeds the threshold, or notifications are silently dropped during an outage.	Search, holds, and hold-notifications are time-sensitive; sustained downtime directly blocks students from core library functions.